Chemicals of Life

Carbohydrates

Proteins

Lipids

Nucleic Acids

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Chemicals of life

These are compounds needed to maintain life of living organisms.

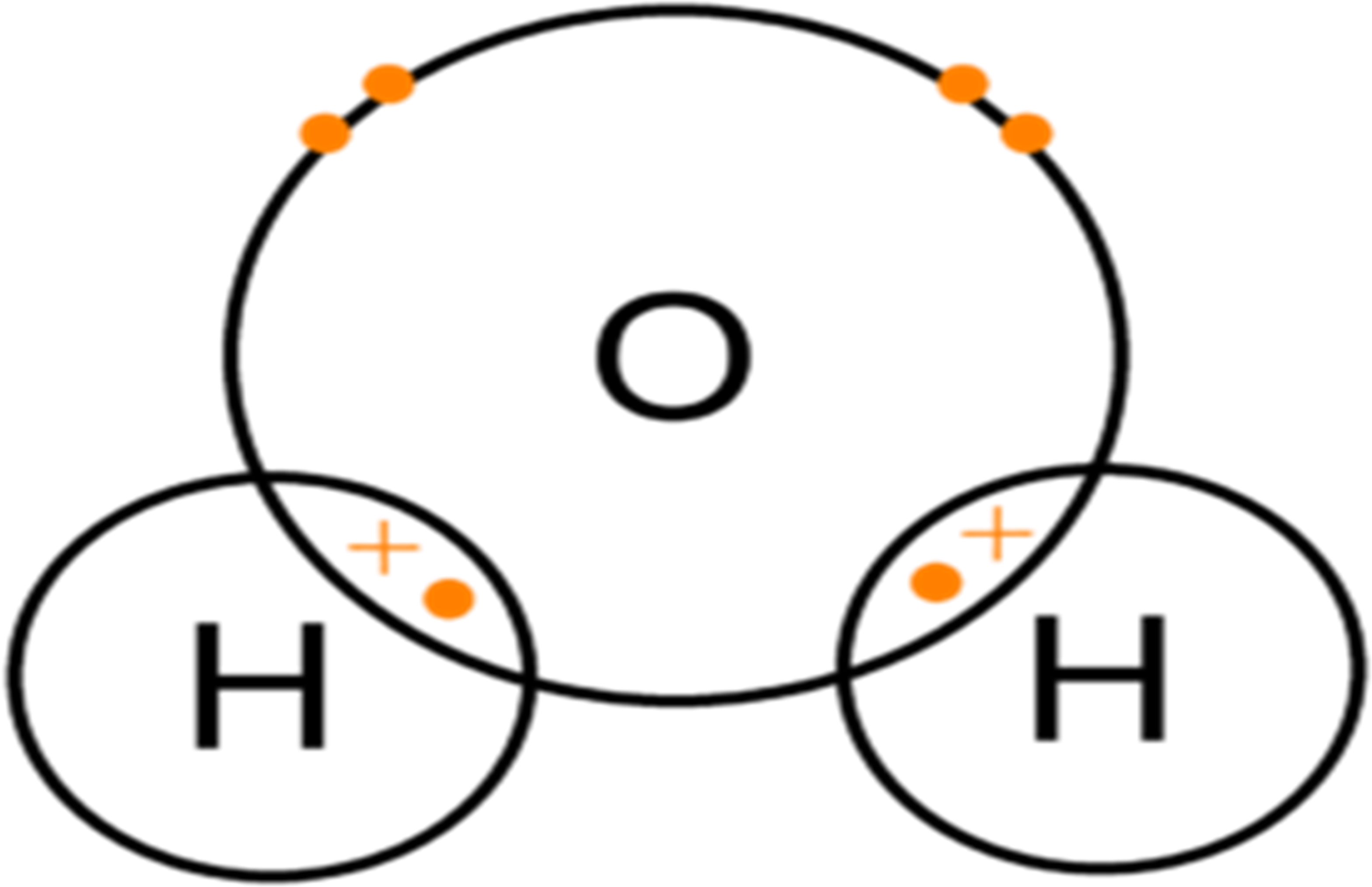
They are divided into two groups, i.e.

- **Inorganic compounds** e.g. water, vitamins, salts, acids and roughages.
- **Organic compounds** e.g. carbohydrates, lipids, proteins and nucleic acids.

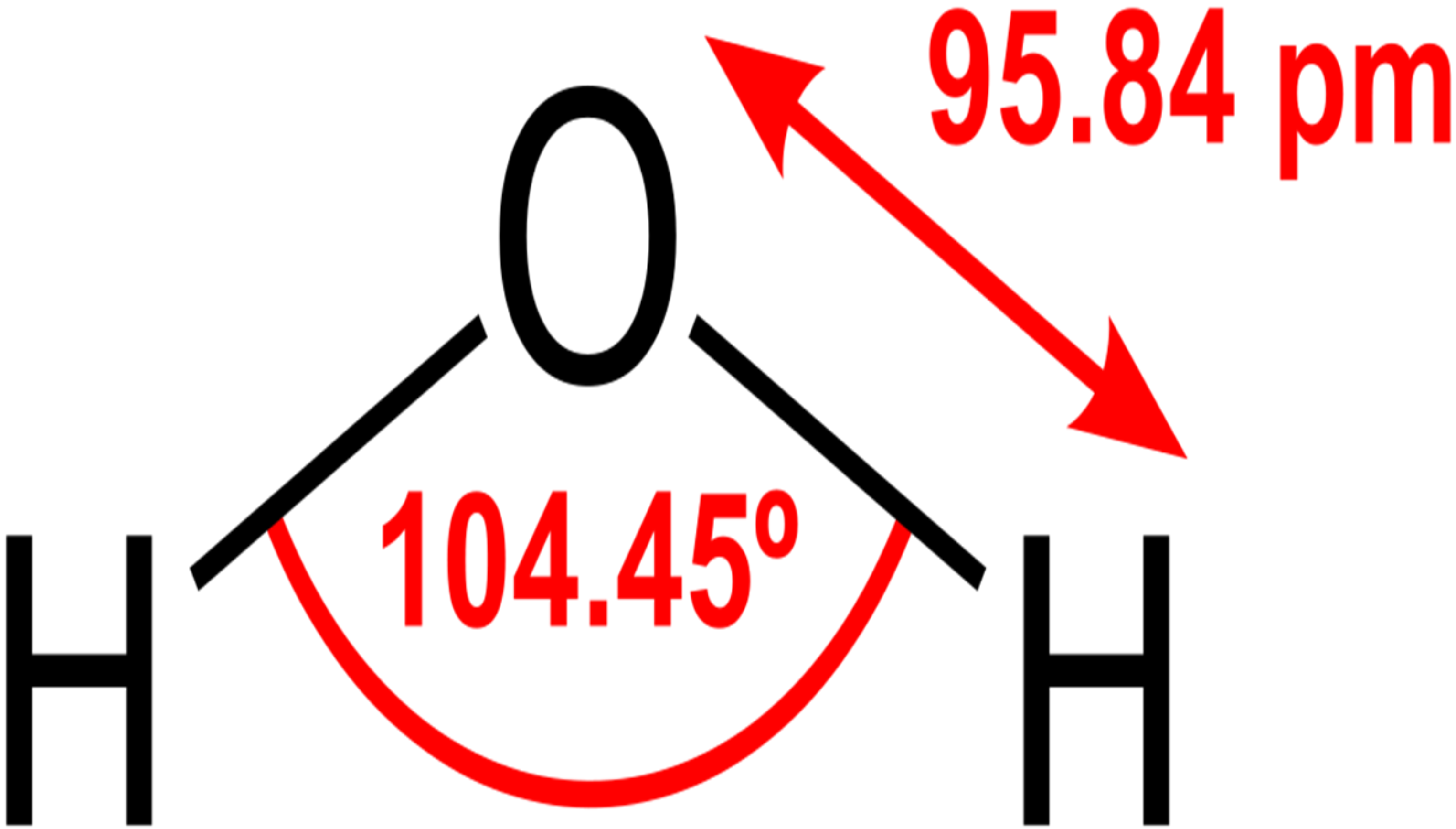
WATER

- A human cell contains about **80%** water and the whole body has over **60%** water.
- Water is formed when two hydrogen atoms combine with an oxygen atom by sharing electrons. The shape of a water molecule is triangular and the angle between the nuclei of atoms is approximately **104^0**
- Water molecules form weak hydrogen bonds with other water molecules nearby and its bonds give it the unique properties.

Structure of water molecule



STRUCTURE OF WATER



Properties of water

i) It is liquid at room temperature.

Properties of water

Learning outcome : 1.1 - WATER b) Describe the properties of water & its importance

2. Low Viscosity

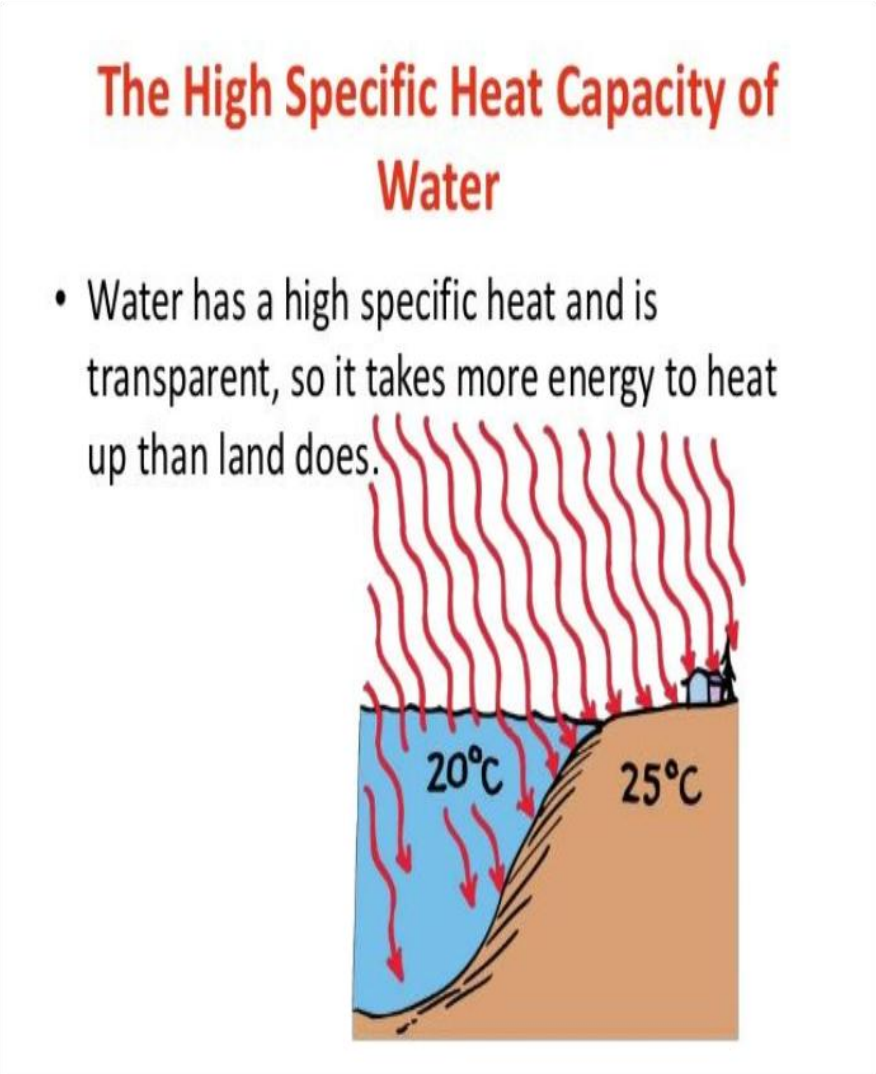
- **Weak hydrogen bonds** between water molecules are constantly break & reform.
- Causing the water molecules slide easily over each other & flow with less friction.
- Thus, water has low viscosity.

Viscosity of Water and Honey

Viscosity of water is considered lower than the honey

Properties of water

iii) It has a high heat capacity therefore much energy is used to raise its temperature because it is used to break the hydrogen bonds which restrict the mobility of the molecules. ***As a result water is relatively slow to heat up or to cool down thus a high heat capacity***



Properties of water

iv). It is the most important inorganic compound in life and most abundant within living organism.

Properties of water

v) Water *expands* as it *freezes* unlike other liquids which contract on cooling.

Properties of water

vi) Water reaches its maximum density above its freezing point at **4°C** hence when water freezes, the ice formed is less dense than the water and hence floats on top of the surface.

In this way, ice insulates water below making it less dense and able to float hence the water will be warmer than the air above.

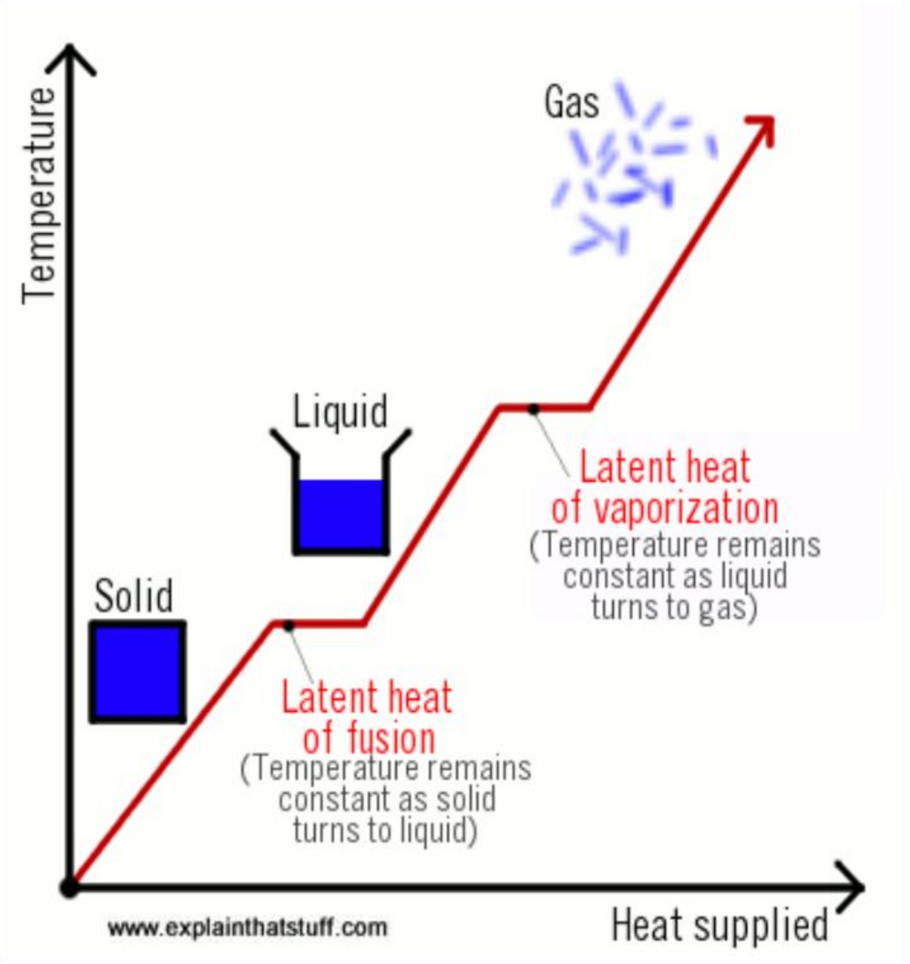
Properties of water

vii) **Water has a high surface tension**. Surface tension is the force that causes the surface of a liquid to contract so that it occupies the least area.

It is high due to the fact that molecules are oriented so that most hydrogen bonds point inwards towards other water molecules.

Properties of water

viii) It has a high latent heat of fusion i.e. much heat must be removed before freezing occurs.



Properties of water

ix) It has high **adhesive** and **cohesive** properties preventing it from breaking under tension.

Properties of water

x) It is colourless and transparent.

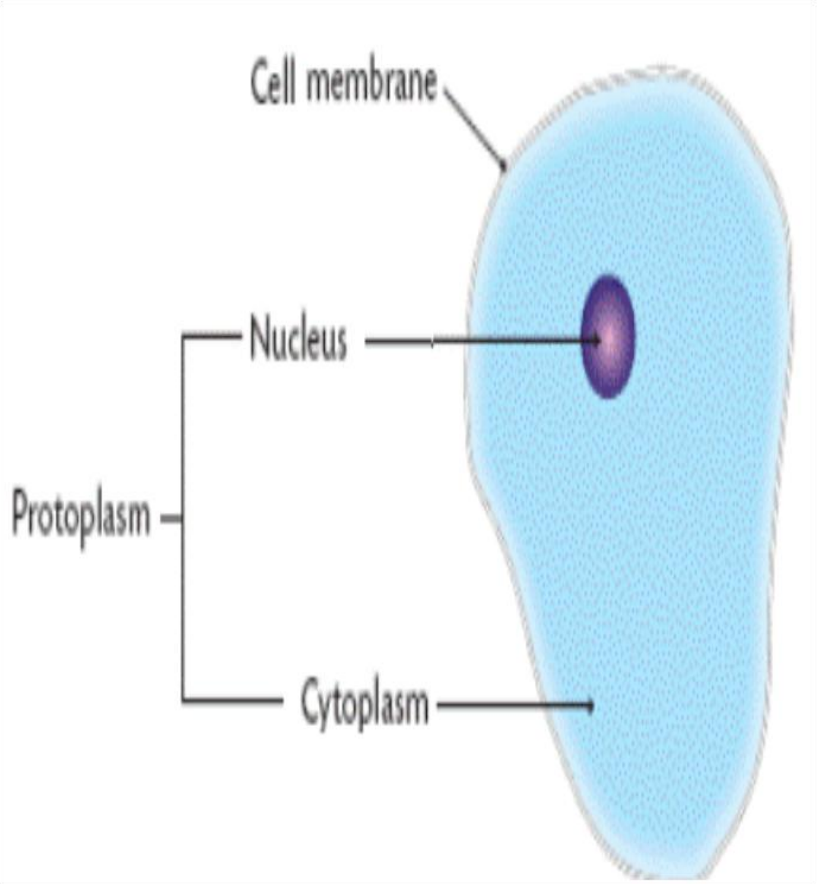
Properties of water

xi) It **dissolves** more substances than any other liquid i.e. it is a ***universal solvent.***

FUNCTIONS OF WATER

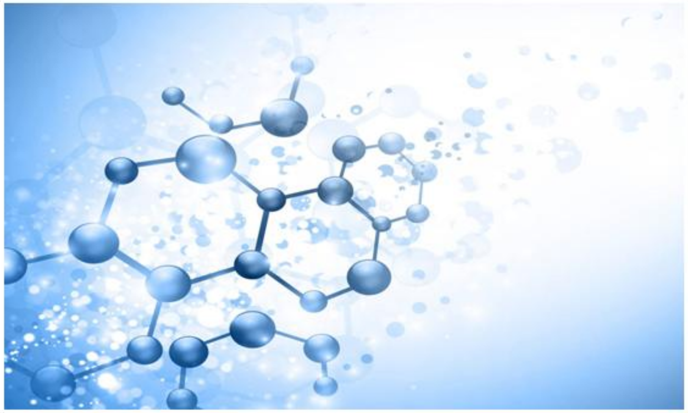
Functions of water

1. It is a ***component*** of **cells**

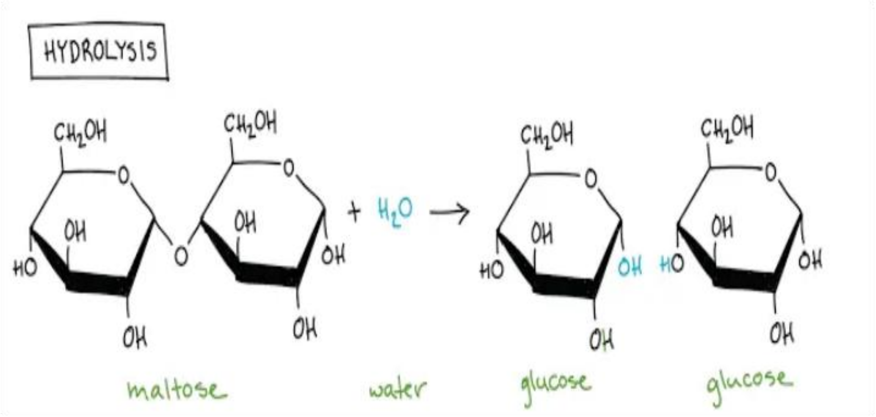


Functions of water

2. It is ***a solvent*** and a medium of transport



3. It is a *reagent* in *hydrolysis*



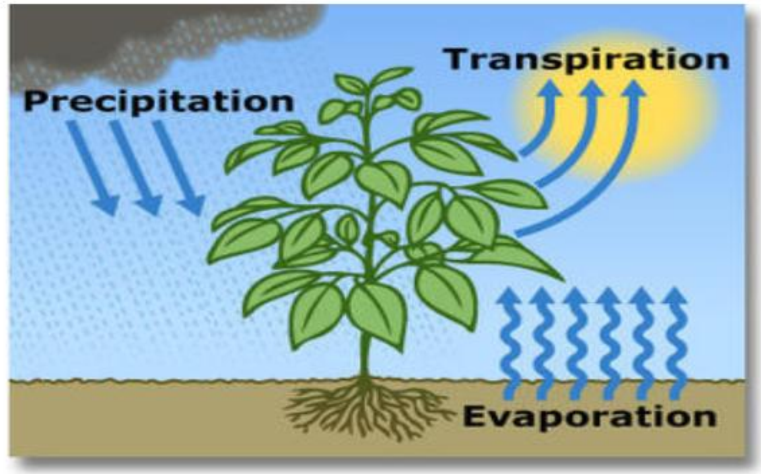
Functions of water

4. It enables ***fertilization*** by swimming gametes

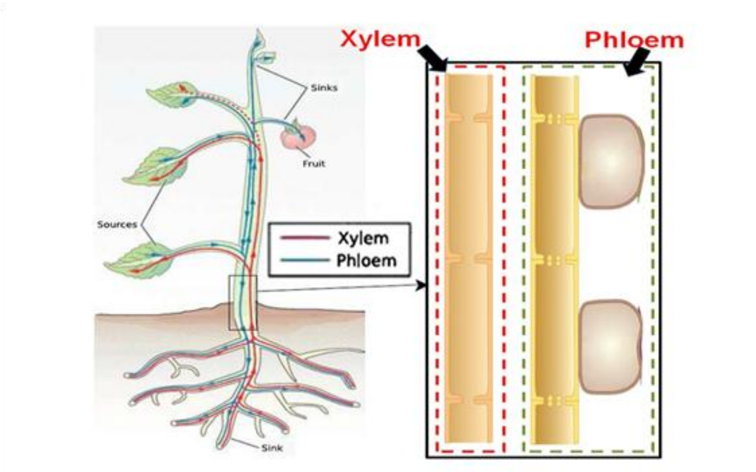
5. It enables ***dispersal*** of seeds, fruits, gametes and larvae stages in aquatic organisms.

Functions of water

6. It is important in ***transpiration*** in plants.



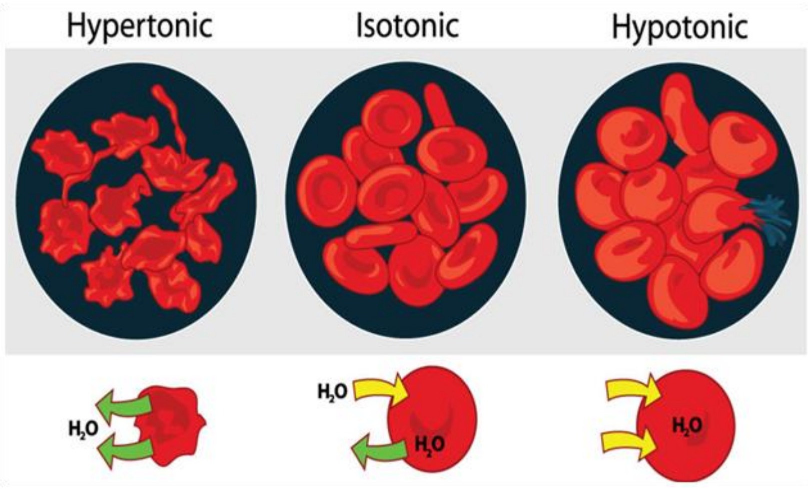
7. It is important in ***translocation*** in plants.



Functions of water

8. It enables ***germination*** to proceed by activating enzymes, transporting hydrolyzed stored food, swelling and breaking open the testa.

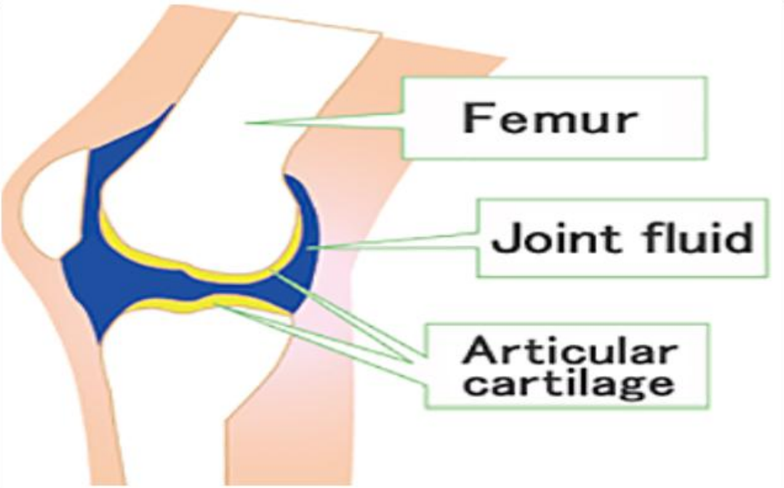
9. It is involved in ***Osmo-regulation*** in animals



Functions of water

10. It ***enables cooling*** by evaporation as a result of sweating and panting.

11. It is a component of **lubricants** at joints e.g. the *synovial fluid*.



Functions of water

12. It offers ***support*** in *hydrostatic skeleton.*

13. It offers ***protection*** as a component of *mucus* and *tears.*

Functions of water

14. It enables ***migration*** to occur as a result of river flow or ocean currents.

QUESTION

How do the properties of water relate to its biological role?

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- ✓ Water is ***transparen***t and this allows light penetration in aquatic habitats to enable **photosynthesis** of aquatic autotrophs and ***visibility*** of aquatic animals.
- ✓ Water has ***a low viscosity*** and this allows for smooth flow of water and other dissolved substances in an aquatic medium for easy transport.
- ✓ It has a ***high surface tension*** providing *support to aquatic organisms* and allowing movement of living organisms on water surface.

CONT…

- ✓ Has ***a high latent heat of vaporization*** hence a cooling effect on the body surface since evaporation of water from the body of an organism draws out excess heat.
- ✓ It has ***a high boiling point*** thus provides **a stable habitat** and medium since a lot of heat which is not normally provided in the natural environment is needed to boil the water.
- ✓ It has ***a high latent heat of fusion*** and hence a low freezing point thus providing a wide range of temperature for survival of aquatic organisms since it prevents freezing of cells and cellular components.

Cont…

- ✓ It has ***a high specific heat capacity*** which minimizes drastic temperature changes in biological systems and provides a constant external environment for many plant cells and aquatic organisms.
- ✓ It has a ***maximum density at 4^{o} C*** hence ice floats on top of water insulating the water below hence **increasing the chances of survival of aquatic organisms below the ice**.

Cont…

- ✓Water is ***liquid at room temperature*** providing a liquid medium for living organisms and metabolic reactions and a medium of transport.
- ✓It has high **adhesive** and **cohesive forces** creating enough capillarity forces for transport in narrow tubes of biological systems.
- ✓It is a **universal solvent** hence providing **a medium for biochemical reactions.**

Cont…

- ✓ Water is **a polar molecule** allowing solubility of polar substances, ionization or dissociation of biochemical substances.
- ✓ Water is **incompressible** thus providing *support in hydrostatic skeleton* and herbaceous stems.
- ✓ Water is **neutral hence** does not alter the pH of cellular components on their environment.
- ✓ A water **molecule** is relatively **small** for easy and fast **transport across a membrane.**